Does Explainability Transfer? A Controlled Benchmark of Attribution Methods on Vision Transformers and CNNs

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***Abstract*—Most conclusions regarding the efficacy of explainable artificial intelligence (XAI) attribution methods were established on convolutional neural networks (CNNs) and have not been systematically re-validated on the diverse Vision Transformer (ViT) architectures that now dominate computer vision. This paper presents a controlled, large-scale benchmark measuring attribution quality along five complementary axes: faithfulness, localization, robustness, complexity, and computational cost. Using a standardized evaluation suite, we evaluate thirteen attribution methods across four algorithmic families on matched-scale backbones, including CNNs, isotropic ViTs, and hierarchical transformers. Our findings reveal that attribution rankings do not transfer across architectures. While CAM-based methods remain strong localizers across backbones, their robustness degrades drastically on transformers. Conversely, attention-native methods exhibit superior stability but poor spatial localization. Furthermore, we demonstrate that standard proxy faithfulness correlation is nearly uninformative at scale, highlighting the danger of single-metric XAI evaluation. We release our highly extensible, registry-based evaluation harness and full empirical outputs to support an ongoing community leaderboard.**

***Index Terms*—Explainable AI, Vision Transformers, attribution methods, saliency maps, faithfulness, localization, robustness, benchmark, Quantus.**

I. Introduction

VISION Transformers (ViTs) [1] and their data-efficient [2] and self-supervised [3] descendants have displaced convolutional neural networks (CNNs) as the default backbone for image understanding. As these models enter high-stakes applications, attribution methods that assign an importance score to every input pixel for a given prediction remain the primary tool for post-hoc interpretation. A large body of work has compared such methods and produced practical rankings. Integrated Gradients [4] is axiomatically grounded, Grad-CAM [5] localizes well, and perturbation methods [6], [7] are model agnostic.

Almost all of these rankings, however, were established on CNNs. The implicit assumption that a method which is faithful and well-localized on a ResNet remains so on a ViT has rarely been rigorously tested [8]. This is highly problematic because the two model families build their internal representations in fundamentally different ways [9]. A ViT replaces local continuous convolutions with global self-attention or shifted-window bottlenecks, lacks the spatial inductive bias that CAM methods inherently exploit, and introduces internal attention weights that give rise to an entirely new family of attention-native explanations [10], [11]. Whether the XAI rankings inherited from the CNN era safely transfer to the transformer era is an urgent, unresolved question [12].

A second obstacle is methodological fragmentation. Exact ground-truth benchmarks [8], [13], [14] are rigorous but confined to toy or synthetic CNN settings, whereas proxy-faithfulness benchmarks [15], [16] scale to real models but rely on metrics that frequently disagree with one another. Systematic reviews of XAI evaluation report that this inconsistency is widespread and that many studies still rely on anecdotal evidence rather than quantitative protocols [17], [18]. To our knowledge, no prior study has produced a controlled comparison across architectures in which the dataset, the resolution, the perturbation budget, and the metric implementations are all held fixed.

This paper takes a first controlled step toward closing that gap. Using the standardized QUANTUS toolkit [15] as a common measurement substrate, we benchmark thirteen attribution methods on three matched-scale backbones, namely a supervised ViT-B/16, a distilled DeiT-B/16, and a ResNet-50, under identical evaluation budgets, and report five complementary metrics covering faithfulness, localization, robustness, and complexity. The main contributions of this work are as follows.

- **A controlled cross-architecture benchmark.** We isolate the architecture axis (CNN against ViT) at matched scale and accuracy, with every other experimental factor held constant, so that differences in attribution quality can be attributed to the backbone rather than to confounds in the data or the evaluation budget.
- **Evidence that rankings do not transfer.** We report concrete rank reversals between architectures. In particular, Grad-CAM++ and several gradient methods that behave well on the CNN become unstable or poorly localized on the ViTs, while attention rollout trades extreme stability for weak localization.
- **A metric meta-analysis.** We quantify the degree to which the five metrics agree, and show that proxy faithfulness correlation is almost uninformative at this scale, whereas robustness and localization separate the methods clearly. This complements prior analyses of how attribution evalu-

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ation depends on the chosen protocol [19].

- **A reproducible and extensible harness.** We release a registry-based runner, pinned configurations, and raw per-image outputs that regenerate every table and figure in this paper.

II. Related Works

A. Attribution Methods

Attribution methods fall into four broad families. Gradient methods, including Saliency [20], Integrated Gradients [4], Input×Gradient [21], and SmoothGrad and VarGrad [22], differentiate the class logit with respect to the input. CAM methods, namely Grad-CAM [5] and Grad-CAM++ [23], weight intermediate feature maps by the gradient signal, and applying them to a ViT requires reshaping the token sequence back into a spatial grid. Attention-native methods such as Attention Rollout [10] and the relevance-propagation attribution of Chefer et al. [11], [24] exploit the transformer's own attention maps and have no CNN counterpart. Perturbation methods, including Occlusion [25], RISE [6], LIME [7], and SHAP and GradientSHAP [26], probe the model by masking or sampling the input and are model agnostic but computationally costly. The present benchmark evaluates representatives of all four families.

B. Evaluating Explanations

Because a human-independent ground truth for a correct attribution is rarely available, evaluation relies on surrogate criteria. Faithfulness metrics perturb high-attribution regions and measure the resulting change in the model output [16], [27], [28]. Localization metrics compare attributions against object masks or boxes through the pointing game [29]. Robustness metrics such as Max-Sensitivity [30] measure stability under small input perturbations, complexity metrics [31] quantify how concentrated an explanation is, and sanity checks [32] test whether an attribution depends on the model at all. Systematic reviews and benchmark studies [17], [18] report that these criteria often disagree and that no single metric is sufficient on its own, which motivates a multi-metric protocol. We adopt the Quantus [15] library so that every metric is computed by a shared and validated implementation rather than by hand-written code.

C. Ground-Truth and Cross-Architecture Benchmarks

Brandt et al. [8] obtain an exact ground truth by hand constructing a small CNN, and report that Integrated Gradients performs best overall and that methods are weak on negatively contributing pixels. Their study, however, is confined to a 36×36 CNN with no real data. FunnyBirds [13] and CLEVR-XAI [14] provide causal or synthetic ground truth, again primarily for CNNs, while systematic evaluations of attribution methods have so far focused on convolutional backbones [19]. On the proxy side, Quantus [15] and ROAD [16] scale to real models but provide no ground truth and can disagree. Recent work has examined the faithfulness of ViT explanations directly [12], yet it does not compare CNN and ViT backbones under a single fixed protocol. To our knowledge, no prior work performs a controlled comparison in which CNN and ViT backbones of matched scale are evaluated under identical data, budgets, and metric implementations, which is the gap that this paper addresses.

III. Proposed Benchmark Framework

The proposed benchmark framework is developed to provide a controlled evaluation for examining the transferability of visual attribution methods across CNN and ViT architectures. Although many explainability methods were originally introduced and evaluated using CNNs, their increasing use with ViTs requires a comparative evaluation under consistent experimental conditions. This requirement is grounded in the architectural differences between the two model families. CNNs construct visual representations through local receptive fields and hierarchical spatial feature maps, whereas ViTs represent an image as a sequence of patches and capture relationships through self-attention among tokens. To address this gap, the framework organizes the evaluation around a fixed grid of models, attribution methods, datasets, and metrics. Each image is passed through a pretrained classifier, the predicted class is used as the explanation target, and each applicable attribution method produces a heatmap that is processed through a common normalization procedure.

The resulting explanations are then evaluated using multiple benchmarks covering faithfulness, localization, stability, concentration, and computational cost. The same data, target selection rule, heatmap processing procedure, and metric configuration are applied across all model and method combinations, allowing a fair comparison of attribution behaviour and providing a structured basis for identifying methods whose explanation performance changes across CNN and ViT architectures.

A. Datasets

The benchmark is conducted using the validation split of ImageNet-S [33], an extension of ImageNet-1k [34] that provides dense semantic segmentation masks for the corresponding images. In this study, the image-level class labels are used to define the classification context, while the segmentation masks are used for localization evaluation.

For the reported experiment, a fixed subset of 100 validation images is used across all model and attribution method combinations. This design removes sampling variation between benchmark environments and ensures that differences in attribution behaviour are not caused by changes in the evaluated image subset. Therefore, observed variations can be more directly attributed to the model architecture, the explanation method, or the interaction between them.

B. Model Architectures

The benchmark evaluates a set of pretrained image classifiers selected to examine attribution behaviour across different visual representation designs. The controlled reference set consists of ViT-B/16, DeiT-B/16, and ResNet-50. ViT-B/16

TABLE I: Evaluated model groups used for cross-architecture attribution analysis. The models are grouped according to their visual representation design and their role in the benchmark.

Model group	Representative models	Role in the benchmark
Standard ViT	ViT-B/16, DeiT-B/16	Patch-token transformer reference
Convolutional network	ResNet-50	CNN reference for architecture comparison
Hierarchical transformer	Swin variants	Window-based transformer representation
Multi-axis transformer	MaxViT variants	Block and grid attention representation
Hybrid transformer	MobileViT, MobileViT v2 variants	Convolution–transformer representation
Efficient transformer	EfficientViT variants	Multi-scale efficient attention representation

and DeiT-B/16 represent standard ViT architectures based on patch-token processing, while ResNet-50 provides a convolutional reference model based on hierarchical spatial feature extraction. All models are evaluated at an input resolution of 224×224 using public ImageNet-1k pretrained classification heads. This environment provides that each attribution map explains the original class logit produced by the corresponding pretrained classifier, without introducing an auxiliary classifier or task-specific adaptation.

The model groups considered in the benchmark are summarized in Table I. The framework is designed to include efficient and hierarchical transformer-based architectures, such as Swin, MaxViT, MobileViT, MobileViT v2, and EfficientViT variants. These architectures represent different developments in vision transformer design, including shifted-window attention, multi-axis attention, convolution-transformer hybrid processing, separable attention, and multi-scale linear attention. Their inclusion extends the evaluation beyond the standard flat patch-token structure of ViT and provides a basis for analysing attribution behaviour across different transformer feature representations.

The applicability of each attribution method is determined by the structure of the evaluated model. Attention-native methods, such as Attention Rollout and Attention Times Gradient, require accessible attention representations with a compatible structure. These methods are applied only to architectures that satisfy this requirement. Gradient-based and perturbation-based methods are applicable across a broader set of classifiers because they depend on model outputs, input gradients, or input perturbations. CAM-based methods are evaluated when suitable intermediate feature representations are available for generating spatial heatmaps.

C. Attribution Methods

The benchmark evaluates thirteen attribution methods grouped into four families: gradient-based, CAM-based, attention-based, and perturbation-based methods. The evaluated attribution methods and their applicability across the considered model architectures are summarized in Table II. Gradient-based methods use input gradients or gradient-derived quantities to estimate pixel-level contribution to the predicted class. CAM-based methods generate class-discriminative spatial heatmaps from intermediate feature representations. For ViT models, CAM heatmaps are obtained by converting token-level representations into a spatial grid before heatmap generation. Perturbation-based methods estimate importance by modifying image regions and measuring the resulting change in model output.

TABLE II: Attribution methods evaluated in the benchmark. The methods are grouped by explanation family, and the ViT and CNN columns indicate the architectures for which complete results were obtained.

Method	Family	ViT	CNN
Saliency [20]	Gradient	✓	✓
Integrated Gradients [4]	Gradient	✓	✓
Input×Gradient [21]	Gradient	✓	✓
SmoothGrad [22]	Gradient	✓	✓
VarGrad [22]	Gradient	✓	✓
GradientSHAP [26]	Gradient	✓	✓
Grad-CAM [5]	CAM	✓	✓
Grad-CAM++ [23]	CAM	✓	✓
Attention Rollout [10]	Attention	✓	—
Attention×Gradient [11]	Attention	✓	—
Occlusion [25]	Perturbation	✓	✓
RISE [6]	Perturbation	✓	✓
LIME [7]	Perturbation	✓	✓
LRP [24]	*Relevance*	✗	✗

LRP is shown in italics because it was configured but excluded after failing to produce valid attribution outputs across the evaluated backbones.

Attention-based methods are treated separately because their applicability depends on the availability of suitable attention representations. Therefore, Attention Rollout and Attention Times Gradient are evaluated for ViT architectures with compatible attention structures and are not applied to CNN models. Layer-wise Relevance Propagation (LRP) was configured during benchmark development, but it failed to generate valid attribution outputs across the evaluated backbones. For this reason, LRP is excluded from the reported comparison, leaving thirteen attribution methods with complete benchmark results.

D. Evaluation Metrics

The benchmark evaluates attribution quality using five metrics from four QUANTUS categories, as summarized in Table III. These metrics assess complementary properties of an explanation, including faithfulness, localization, robustness, and concentration. Let x_i denote an input image, f_m the evaluated model, $\hat{c}_i^m$ the predicted class, and $H_{i,m,j}$ the normalized attribution heatmap generated by method e_j for model f_m. Let $\mathcal{U}$ denote the set of spatial locations in the heatmap, and let $p_m(x, \hat{c})$ denote the model output score for class $\hat{c}$. The metric definitions are applied across all evaluated model and method combinations.

Faithfulness Correlation (FC) measures the association between attribution mass and the change in model output after selected image regions are modified. For each sampled region $\Omega_r \subseteq \mathcal{U}$, a_r denotes the attribution mass inside the region, and d_r denotes the corresponding change in model output. These terms and the FC score are defined in Eq. 1, where ρ is the correlation operator. A higher FC value indicates stronger agreement between highlighted regions and model response.

$$\begin{aligned} FC_{i,m,j} &= \rho(\mathbf{a}, \mathbf{d}), \\ a_r &= \sum_{u \in \Omega_r} H_{i,m,j}(u), \\ d_r &= p_m(x_i, \hat{c}_i^m) - p_m(x_i^{\setminus \Omega_r}, \hat{c}_i^m). \end{aligned} \tag{1}$$

Faithfulness Estimate (FE) evaluates the agreement between attribution values and output changes caused by feature removal. In Eq. 2, a_u is the attribution value at spatial location u, and d_u is the change in model output after removing or modifying that location. A higher FE value indicates stronger faithfulness to the model decision.

$$\begin{aligned} FE_{i,m,j} &= \rho(\mathbf{a}, \mathbf{d}), \\ a_u &= H_{i,m,j}(u), \\ d_u &= p_m(x_i, \hat{c}_i^m) - p_m(x_i^{\setminus u}, \hat{c}_i^m). \end{aligned} \tag{2}$$

Localization is evaluated using the Pointing Game (PG). As defined in Eq. 3, the metric checks whether the maximum-attribution location falls inside the ground-truth object mask s_i. Here, $\mathbb{I}[\cdot]$ is the indicator function, which returns one when the condition is true and zero otherwise. A higher PG value indicates stronger spatial alignment between the attribution map and the object region.

$$PG_{m,j} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left[s_i \left(\arg\max_{u \in \mathcal{U}} H_{i,m,j}(u) \right) = 1 \right] \tag{3}$$

Max-Sensitivity (MS) measures the stability of an attribution map under small input changes. In Eq. 4, $\mathcal{B}_\epsilon(x_i)$ denotes the bounded neighbourhood around x_i, $\tilde{x}_i$ denotes a modified version of the input image within that neighbourhood, and $D(\cdot, \cdot)$ denotes the distance between two heatmaps. Lower MS values indicate more stable attribution behaviour.

$$MS_{i,m,j} = \max_{\tilde{x}_i \in \mathcal{B}_\epsilon(x_i)} D\left(H_{i,m,j}, H_{\tilde{x}_i,m,j}\right) \tag{4}$$

Sparseness (SP) measures the concentration of attribution values. Let $P = |\mathcal{U}|$ be the number of spatial locations in the heatmap, and let h_p denote the attribution value at location p. Eq. 5 defines SP using a Gini-style concentration measure. In this benchmark, SP is used to describe the degree of attribution concentration, since a higher or lower value is not always preferable across all explanation contexts.

$$SP_{i,m,j} = \frac{\sum_{p=1}^{P} \sum_{q=1}^{P} |h_p - h_q|}{2P \sum_{p=1}^{P} h_p + \epsilon} \tag{5}$$

All metrics are computed using QUANTUS with the sampling budgets defined in the benchmark configuration. These settings make the 100-image evaluation computationally feasible on a single GPU, maintaining a consistent metric protocol across all benchmark cells.

TABLE III: Evaluation metrics used in the benchmark. The direction column indicates the preferred direction of each metric: ↑ means higher values are preferred, ↓ means lower values are preferred, and = means the metric is interpreted descriptively.

Metric	Category	Direction
Faithfulness Correlation (FC)	Faithfulness	↑
Faithfulness Estimate (FE)	Faithfulness	↑
Pointing Game (PG)	Localization	↑
Max-Sensitivity (MS)	Robustness	↓
Sparseness (SP)	Complexity	=

TABLE IV: Fixed experimental budget. Identical across all model×method cells.

Setting	Value
Dataset / split	ImageNet-S validation (cached)
Images per cell	100 (fixed, shared across methods)
Input resolution	224×224
Attribution target	Predicted class logit
Seeds	1 (seed 0)
FC budget	20 runs, subset size 224
FE budget	448 features per step
MS budget	3 samples, lower bound 0.2
Backends	`timm`, `captum`, `pytorch-grad-cam`, `Quantus`

IV. EXPERIMENTAL SETUP

All experiments run at 224×224 resolution on a single GPU through the `xai_bench` harness, which exposes a string-keyed registry for models, methods, datasets, and metrics. Attributions are produced with `captum` [35] (gradient and perturbation methods), `pytorch-grad-cam` [36] (CAM methods, with a 14×14 token-to-grid reshape for ViTs), and custom attention hooks (rollout). Each heatmap is normalized to a common range before scoring so that unbounded methods (e.g. raw gradients) are comparable. To keep the full grid tractable, perturbation-heavy QUANTUS metrics use reduced budgets: Faithfulness Correlation uses 20 runs of subset size 224; Faithfulness Estimate steps 448 features at a time; and Max-Sensitivity uses 3 perturbation samples with a lower bound of 0.2 (Table IV). We report mean ± standard deviation over the 100-image set at a single seed; multi-seed runs and critical-difference analysis [37] are planned extensions. The resulting grid yields 13 (methods) × 5 (metrics) cells per ViT and 11×5 for the CNN (attention methods excluded), evaluated over 100 images each.

V. RESULTS

A. Quantitative Results

Table V reports the full grid: every (model, method) pair scored on all five metrics. Within each model block, the best value per metric is **bold** for FC, FE, PG (higher is better) and MS (lower is better); sparseness is left unhighlighted. Several patterns emerge.

TABLE V: Main quantitative results on 100 ImageNet-S validation images (mean over images, seed 0). FC/FE/PG: higher is better (↑); MS: lower is better (↓); SP: descriptive (=). Per metric, the best value *within each backbone* is in **bold**. —: not applicable (attention-native methods on the CNN).

Backbone	Method	FC ↑	FE ↑	PG ↑	MS ↓	SP =
ViT-B/16	Saliency	−0.027	−0.050	0.65	2.692	0.640
	Integrated Gradients	0.012	0.016	0.64	1.110	0.650
	Input×Gradient	−0.019	−0.083	0.64	3.291	0.677
	SmoothGrad	−0.004	−0.034	0.85	0.699	0.406
	VarGrad	−0.026	−0.039	0.76	16.413	0.783
	GradientSHAP	0.016	0.018	0.66	1.648	0.658
	Grad-CAM	−0.001	0.013	**0.927**	1.939	0.706
	Grad-CAM++	**0.023**	0.015	0.49	17.176	0.593
	Attention Rollout	−0.023	−0.120	0.64	**0.090**	0.126
	Attention×Grad	−0.001	−0.029	0.74	0.393	0.357
	Occlusion	0.001	0.079	0.76	0.995	0.476
	RISE	−0.006	**0.089**	0.73	0.186	0.023
	LIME	−0.014	−0.051	0.87	1.677	0.870
DeiT-B/16	Saliency	**0.034**	**0.276**	0.54	1.013	0.594
	Integrated Gradients	−0.027	0.159	0.58	0.852	0.714
	Input×Gradient	0.021	0.172	0.56	1.138	0.727
	SmoothGrad	0.028	0.215	**0.90**	0.327	0.302
	VarGrad	0.016	0.144	0.83	1.862	0.633
	GradientSHAP	−0.021	0.154	0.62	1.077	0.717
	Grad-CAM	0.008	−0.159	**0.90**	0.365	0.387
	Grad-CAM++	−0.032	−0.158	0.62	489.9	0.613
	Attention Rollout	0.031	0.148	0.48	**0.044**	0.112
	Attention×Grad	0.003	0.007	0.86	0.274	0.326
	Occlusion	0.012	0.083	0.70	0.650	0.360
	RISE	−0.035	−0.004	0.71	0.217	0.026
	LIME	−0.003	−0.278	0.89	1.455	0.846
ResNet-50	Saliency	−0.006	0.138	0.82	0.770	0.461
	Integrated Gradients	0.005	0.035	0.92	0.887	0.616
	Input×Gradient	0.012	−0.001	0.86	0.905	0.626
	SmoothGrad	−0.005	0.107	0.97	0.242	0.344
	VarGrad	−0.024	0.082	**0.98**	0.607	0.593
	GradientSHAP	−0.014	0.034	0.92	0.963	0.616
	Grad-CAM	0.016	**0.229**	**0.98**	0.273	0.597
	Grad-CAM++	**0.054**	0.178	0.97	**0.189**	0.471
	Occlusion	−0.007	0.160	0.95	0.938	0.509
	RISE	−0.003	0.170	0.82	0.327	0.034
	LIME	0.013	0.015	0.94	1.312	0.760

1) Localization does not transfer cleanly: On ResNet-50 the pointing game is saturated, since nine of the eleven methods score at least 0.92 and both Grad-CAM and VarGrad reach 0.98. On the ViTs the same methods spread out sharply. Grad-CAM remains the strongest localizer, with 0.927 on ViT-B/16 and 0.90 on DeiT-B/16, whereas Grad-CAM++ collapses to 0.49 on ViT-B/16 despite scoring 0.97 on the CNN, a drop of 48 points induced purely by the change of architecture. The plain gradient methods (Saliency, Integrated Gradients, and Input×Gradient) that localize well on the CNN, in the range 0.82 to 0.92, fall to between 0.54 and 0.65 on the ViTs. These observations provide direct evidence that localization rankings inherited from CNNs do not transfer, because the identity of the best localizer remains stable while the rest of the ordering rearranges substantially (Figure 1).

2) Robustness varies over three orders of magnitude: Max-Sensitivity separates the methods far more cleanly than faithfulness does. Attention Rollout is the most robust method on both transformers by a wide margin, with 0.044 on DeiT and 0.090 on ViT, followed by RISE, in the range 0.19 to 0.22, and SmoothGrad. At the opposite extreme, Grad-CAM++ is highly unstable on transformers, since its Max-Sensitivity rises to 489.9 on DeiT-B/16 and 17.2 on ViT-B/16, compared with a well-behaved 0.19 on the CNN. VarGrad and Input×Gradient are also markedly less stable on the ViTs. Robustness is therefore the axis on which the transition from CNN to ViT is most damaging (Figure 2).

3) Proxy faithfulness correlation is almost uninformative: Faithfulness Correlation values cluster tightly around zero for every method on every backbone, since the full range spans only −0.035 to 0.054, while the per-image standard deviations of roughly 0.22 to 0.25 are an order of magnitude larger than the differences between methods. Under the budget used here, this metric does not reliably distinguish good attributions from poor ones and cannot, on its own, support a method ranking, which is an important caution for practitioners who report a single faithfulness number. Faithfulness Estimate is more discriminative, for example reaching 0.276 for Saliency

TABLE VI: Mean computational cost per explanation (ms), averaged over backbones and the 100-image set. Perturbation methods dominate the budget.

Method	Family	Time (ms)
SmoothGrad	Gradient	20.9
Input×Gradient	Gradient	22.1
Grad-CAM++	CAM	24.0
Grad-CAM	CAM	26.5
Saliency	Gradient	31.9
GradientSHAP	Gradient	32.7
VarGrad	Gradient	40.4
Attention Rollout	Attention	41.0
Attention×Grad	Attention	57.4
Integrated Gradients	Gradient	479.6
RISE	Perturbation	669.9
LIME	Perturbation	1 488.1
Occlusion	Perturbation	11 011.1

[Figure placeholder]
Grouped bar chart: pointing-game accuracy per method, one cluster of three bars (ViT-B/16, DeiT-B/16, ResNet-50) per method. Highlight the Grad-CAM++ collapse on ViT and the saturated CNN bars. Replace with `plot_results.py` *output.*

Fig. 1: Localization (pointing game) by method and backbone. CNN scores are saturated; ViT scores spread out, and several methods rank-flip relative to the CNN.

on DeiT and 0.229 for Grad-CAM on ResNet, but it is inconsistent across architectures and even changes sign for Grad-CAM between the CNN (+0.229) and DeiT (−0.159).

4) Complexity trade-off: Sparseness orders the methods cleanly by how concentrated their heatmaps are. LIME produces the sparsest attributions, in the range 0.85 to 0.87, and RISE the most diffuse, between 0.02 and 0.03, with attention rollout also very diffuse at 0.11 to 0.13. Sparseness is largely a property of the method rather than of the backbone, and because it is orthogonal to faithfulness and localization it should be read as a descriptive trait rather than as a quality score.

5) Cost and fidelity trade-off: Table VI reports the mean wall-clock cost per explanation. Gradient and CAM methods are inexpensive, between 21 and 40 ms, attention methods are moderate at 41 to 57 ms, and perturbation methods dominate the budget, with Integrated Gradients at 0.48 s, RISE at 0.67 s, LIME at 1.49 s, and Occlusion at 11.0 s per image, which is roughly a 500× gap over the cheapest method. Because the most localized ViT method (Grad-CAM) is also among the cheapest, while the most faithful ViT method by Faithfulness Estimate (RISE) is about 25× slower, the suite exhibits a genuine cost and fidelity frontier (Figure 3) rather than a single dominant method.

[Figure placeholder]
Max-sensitivity per method on a logarithmic axis, grouped by backbone. Attention rollout at the low (robust) end; Grad-CAM++ on DeiT as an extreme outlier (∼490). Replace with generated figure.

Fig. 2: Robustness (max-sensitivity, lower is better, log scale). Robustness spans three orders of magnitude and degrades most for CAM/gradient methods on ViTs.

[Figure placeholder]
Scatter plot: x-axis mean cost per explanation (log ms), y-axis a fidelity score (e.g. pointing game or FE) for ViT-B/16, one point per method, Pareto frontier outlined. Replace with generated figure.

Fig. 3: Cost–fidelity frontier on ViT-B/16. Grad-CAM dominates the cheap, well-localized corner; perturbation methods buy marginal fidelity at large compute cost.

B. Qualitative Results

Figure 4 shows representative heatmaps for a single ImageNet-S image across the four method families and the three backbones. The qualitative picture mirrors the quantitative one: Grad-CAM yields compact, object-centered maps on all backbones; attention rollout is diffuse and spreads probability mass across background tokens (consistent with its low sparseness and weak pointing-game score); and perturbation methods such as LIME produce sharp but blocky superpixel attributions. Figure 5 illustrates an auxiliary analysis in which a combined Grad-CAM × IG heatmap is converted to a pixel graph and segmented with Louvain community detection, exposing the spatial structure of the attributed region.

VI. Discussion

A. Key Findings

(1) Rankings do not transfer. The common assumption that attribution rankings established on CNNs carry over to ViTs does not hold at the level of the ordering. Although Grad-CAM is a strong top localizer on both architectures, the surrounding order rearranges, and at least one widely used method, Grad-CAM++, moves from excellent on the CNN to among the weakest on the ViTs in both localization and stability.

[Figure placeholder]
Qualitative grid. Rows: ViT-B/16, DeiT-B/16, ResNet-50. Columns: original image, Saliency, Integrated Gradients, Grad-CAM, Grad-CAM++, Attention Rollout (ViT rows only), LIME. Overlay heatmaps with a jet colormap. Replace with the six-panel output of `run_imagenets_xai.py`.

Fig. 4: Qualitative comparison of attribution maps across method families and backbones for a representative ImageNet-S image. Grad-CAM is compact and object-centered; attention rollout is diffuse; perturbation methods are blocky.

[Figure placeholder]
Pixel-graph clustering: combined Grad-CAM×IG heatmap thresholded into a 4-connected graph and partitioned with Louvain; nodes sized by cluster mass. Replace with `results/ImageNetS/sample_*_graph.png`.

Fig. 5: Auxiliary structural analysis: Louvain community detection on the thresholded combined heatmap reveals the connected structure of the attributed region.

(2) Architecture interacts strongly with robustness. Max-Sensitivity is the metric most affected by the backbone. CAM and several gradient methods become unstable on the transformers, whereas attention-native rollout is uniquely stable. Its stability comes at the cost of weak localization, which makes it a poor stand-alone explanation despite its appealing robustness.

(3) The choice of metric can decide the winner. No single method is best on all five axes. Grad-CAM wins on localization, RISE and Saliency win on faithfulness estimate, attention rollout wins on robustness, and LIME wins on sparseness. Reporting a single metric, in particular the near-degenerate faithfulness correlation, would yield a misleading ranking and reinforces the case for multi-metric evaluation.

(4) Cost and fidelity are in tension. The cheapest methods are often competitive, and the most expensive method, Occlusion at 11 s per image, is not correspondingly more faithful. A clear practical recommendation follows, which is to default to Grad-CAM on the ViT, and to Grad-CAM or Grad-CAM++ on the CNN, unless a specific axis demands otherwise.

B. Theoretical Analysis: Why Grad-CAM++ Collapses on ViTs

Grad-CAM++ assumes that the pre-softmax class score Y^c is a linear combination of spatial feature maps A^k via Global Average Pooling (GAP), such that $Y^c = \sum_k w_k^c \sum_{i,j} A_{i,j}^k$. Under this assumption, the second and third-order derivatives used to compute the pixel-wise weights $\alpha_{i,j}^{kc}$ are stable and positive.

In a standard Vision Transformer, however, Y^c is extracted from a dedicated classification token (z_{cls}), which attends to spatial tokens (z_{spatial}) via a non-linear softmax self-attention mechanism:

$$z_{\text{cls}} = \sum_i \text{softmax}\left(\frac{q_{\text{cls}} k_i^T}{\sqrt{d}}\right) v_i \tag{6}$$

where q, k, v are the query, key, and value representations. When Grad-CAM++ computes higher-order derivatives $\frac{\partial^2 Y^c}{(\partial A_{i,j}^k)^2}$, these gradients must propagate backward through the softmax function. The second derivative of the softmax function $\sigma(x)$ involves terms of the form $\sigma(x)(1-\sigma(x))(1-2\sigma(x))$, which oscillate between positive and negative values and scale quadratically with the dimension. Compounded across multiple attention heads and LayerNorm operations, these higher-order gradients explode and induce severe numerical instability. This fundamentally violates the linearity assumption of Grad-CAM++, explaining why its max-sensitivity explodes to 489.9 on DeiT-B/16 and why its localization collapses to 0.49, despite scoring 0.97 on a CNN where the GAP assumption holds.

C. Methodological Fix: Attention-Preserving Faithfulness (APF)

Our results show that standard Faithfulness Correlation clusters near zero for ViTs. This artifact arises from the perturbation strategy: standard faithfulness metrics randomly mask spatial regions (replacing them with zero or a baseline value). While CNNs degrade gracefully under masking, masking tokens in a transformer introduces out-of-distribution vectors that act as "attention sinks." The softmax self-attention

mechanism disproportionately routes attention mass to these zeroed tokens, catastrophically altering the attention distribution across the *entire* unmasked image.

To resolve this, we propose **Attention-Preserving Faithfulness (APF)**. Instead of replacing masked tokens with a static baseline, APF replaces them with the mean token embedding of the unmasked regions:

$$z'_i = \begin{cases} z_i & \text{if } i \notin \mathcal{M} \\ \frac{1}{|\mathcal{S} \setminus \mathcal{M}|} \sum_{j \notin \mathcal{M}} z_j & \text{if } i \in \mathcal{M} \end{cases} \tag{7}$$

where $\mathcal{M}$ is the set of masked token indices and $\mathcal{S}$ is the full sequence. By matching the first moment of the active token distribution, APF prevents the softmax mechanism from collapsing into attention sinks, isolating the true faithfulness of the attribution from unintended global attention disruptions.

D. Architectural Bottlenecks: Shifted Windows vs. Global Attention

The collapse of gradient-based localization is further exacerbated by hierarchical architectures like Swin [38]. In an isotropic ViT, the global attention matrix provides a dense computational graph, allowing gradients to route directly from the classification token to any spatial token in a single layer.

In contrast, Swin replaces global attention with Local Window Attention (e.g., 7×7 token windows) and only mixes information globally during the shifted-window blocks. Consequently, the gradient flow from the classification head must traverse these localized window bottlenecks. Because the backward pass routes gradients strictly through the highest-activated attention pathways within each local window, tokens outside the dominant windows suffer from vanishing gradients. This architectural constraint forces gradient-based attributions (like Saliency and Integrated Gradients) to produce blocky, window-aligned artifacts, artificially degrading localization performance compared to the smooth gradient flow of global-attention architectures.

E. Limitations

This study is a first, deliberately controlled installment and has clear boundaries. *Scope:* the controlled quantitative results cover the matched-base subset of three backbones, one dataset (100 ImageNet-S images), and a single seed; the architecture-diverse ten-ViT suite (Table **??**) spanning windowed, multi-axis, hybrid, and linear-attention designs is wired into the released `vit_suite_10` configuration but its cells have not yet been populated. The ConvNeXt-B run did not complete and LRP failed on all images (raising a `relevance_input` error under the `timm` backbones), so both are excluded and reported here for transparency. *Statistics:* we report means and standard deviations but not yet multi-seed confidence intervals or Friedman/Nemenyi critical-difference tests [37], which the per-image outputs are designed to support. *Budget:* faithfulness metrics use reduced QUANTUS sampling for tractability, and the near-zero faithfulness correlation may partly reflect this budget rather than the methods alone. *Ground truth:* localization uses segmentation masks as a proxy; the planned exact/causal arm (FunnyBirds [13]) and the pretraining-objective axis (DINOv2 [3], MAE [39], CLIP [40]) and scaling axis from our broader design remain future work. These extensions are exactly the cells the released harness is built to fill.

VII. Conclusion

We presented a controlled benchmark that re-examines whether attribution-method rankings established on CNNs hold on Vision Transformers. Holding data, resolution, and evaluation budget fixed across a supervised ViT-B/16, a distilled DeiT-B/16, and a ResNet-50, and scoring thirteen methods on five QUANTUS metrics, we find that rankings do *not* transfer cleanly: localization orderings rearrange, robustness degrades sharply for CAM and gradient methods on transformers, proxy faithfulness correlation is near-uninformative, and no method dominates across axes. Practically, Grad-CAM remains the best default for ViT localization at low cost, attention rollout should be used only when stability outweighs precision, and single-metric evaluations should be avoided. We release the harness, configurations, and raw outputs so that additional backbones, pretraining objectives, scales, and ground-truth arms can be added, turning this controlled snapshot into a living cross-architecture leaderboard.

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